Lesson 1

Silvia Giulia Galfrè

Introduction to the course

About me

Silvia Giulia Galfrè

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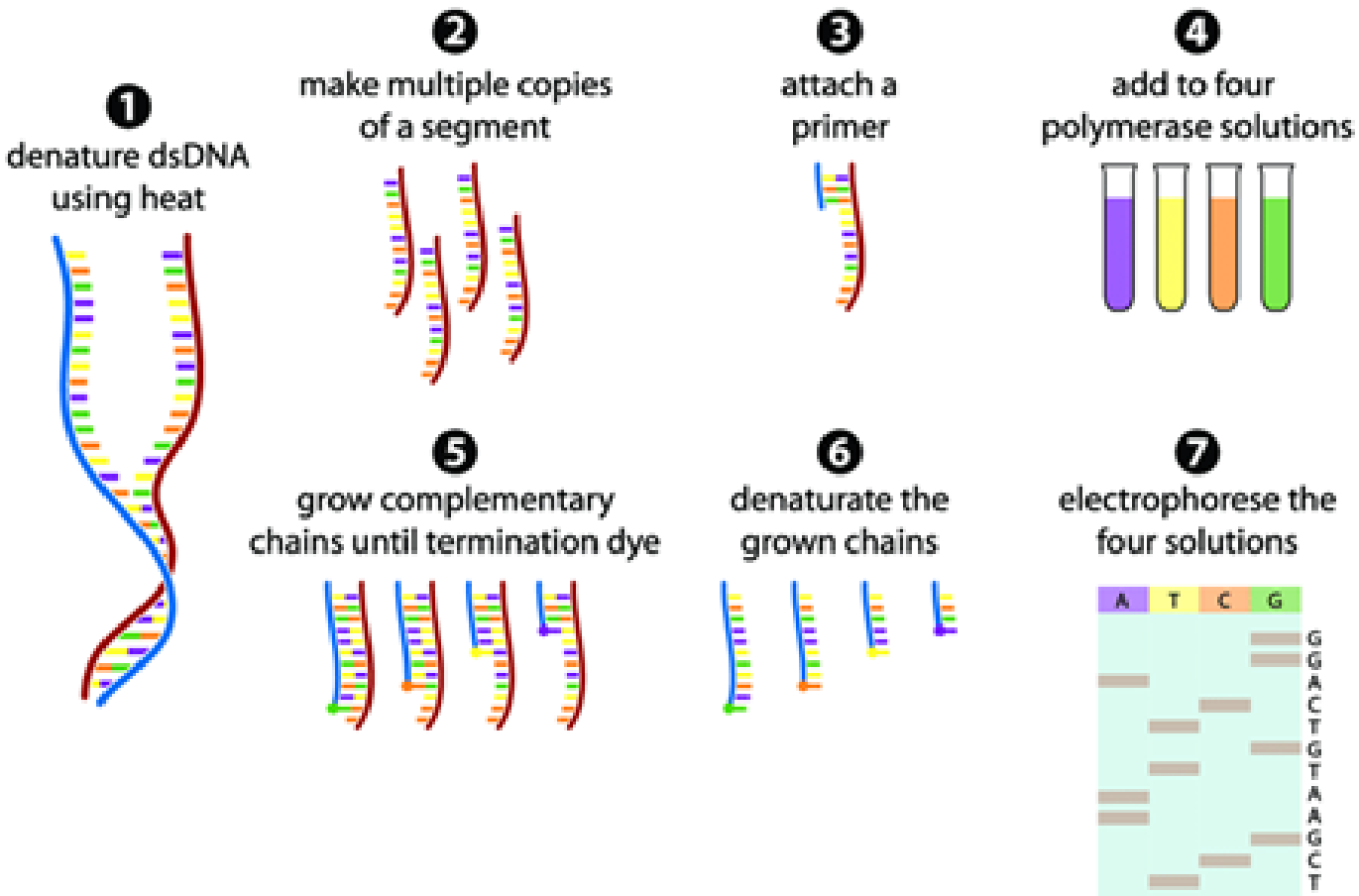
Office hours: by appointment

Research: computational biology & bioinformatics, transcriptomics, data science, applied machine learning

From sequencing to data

THE 3 GENERATIONS OF SEQUENCING PLATFORMS

Sanger sequencing: first generation



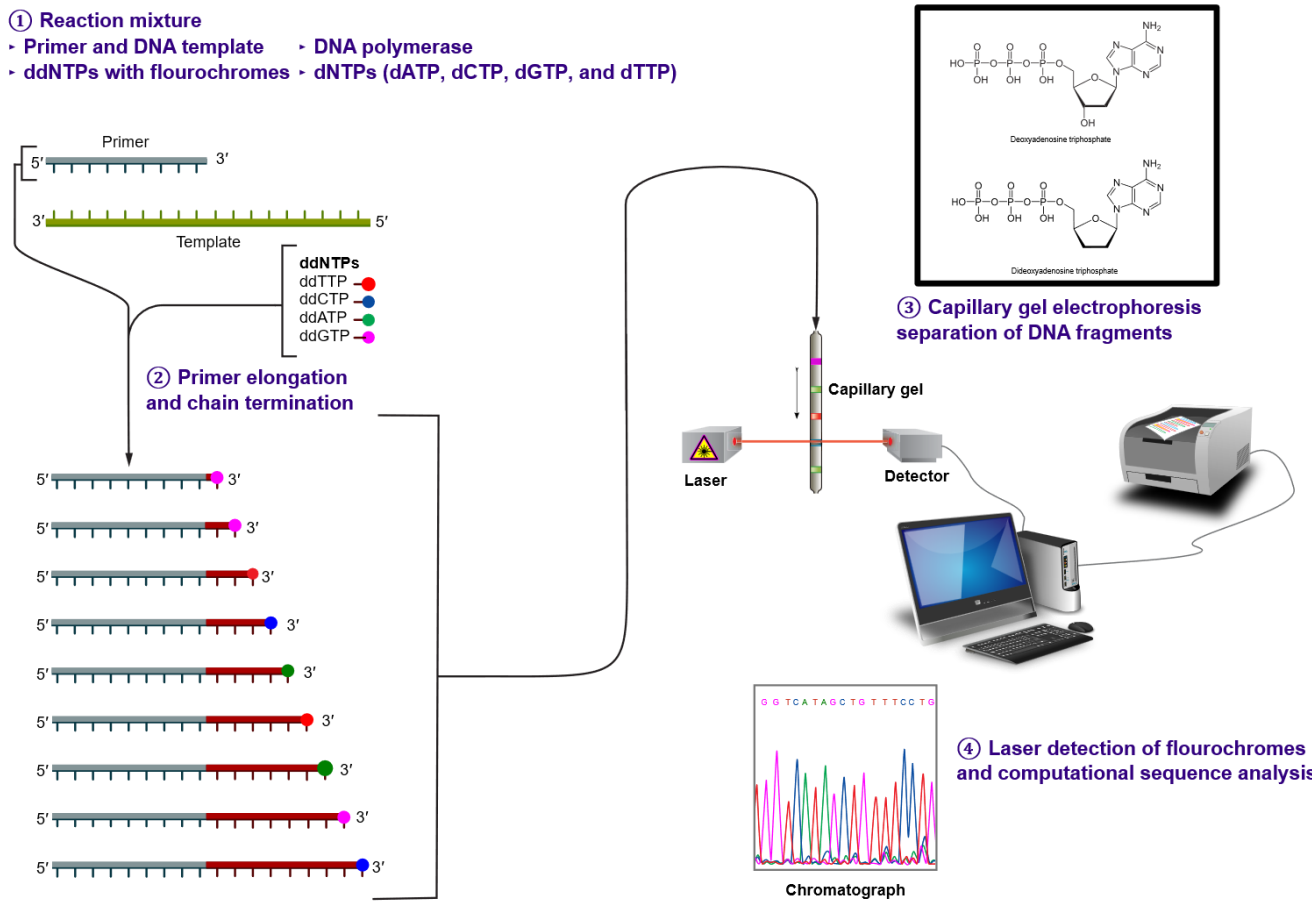
https://frontlinegenomics.com/a-history-of-sequencing/

Sanger sequencing: first generation

After initial gel electrophoresis
A big improvement: capillary electrophoresis

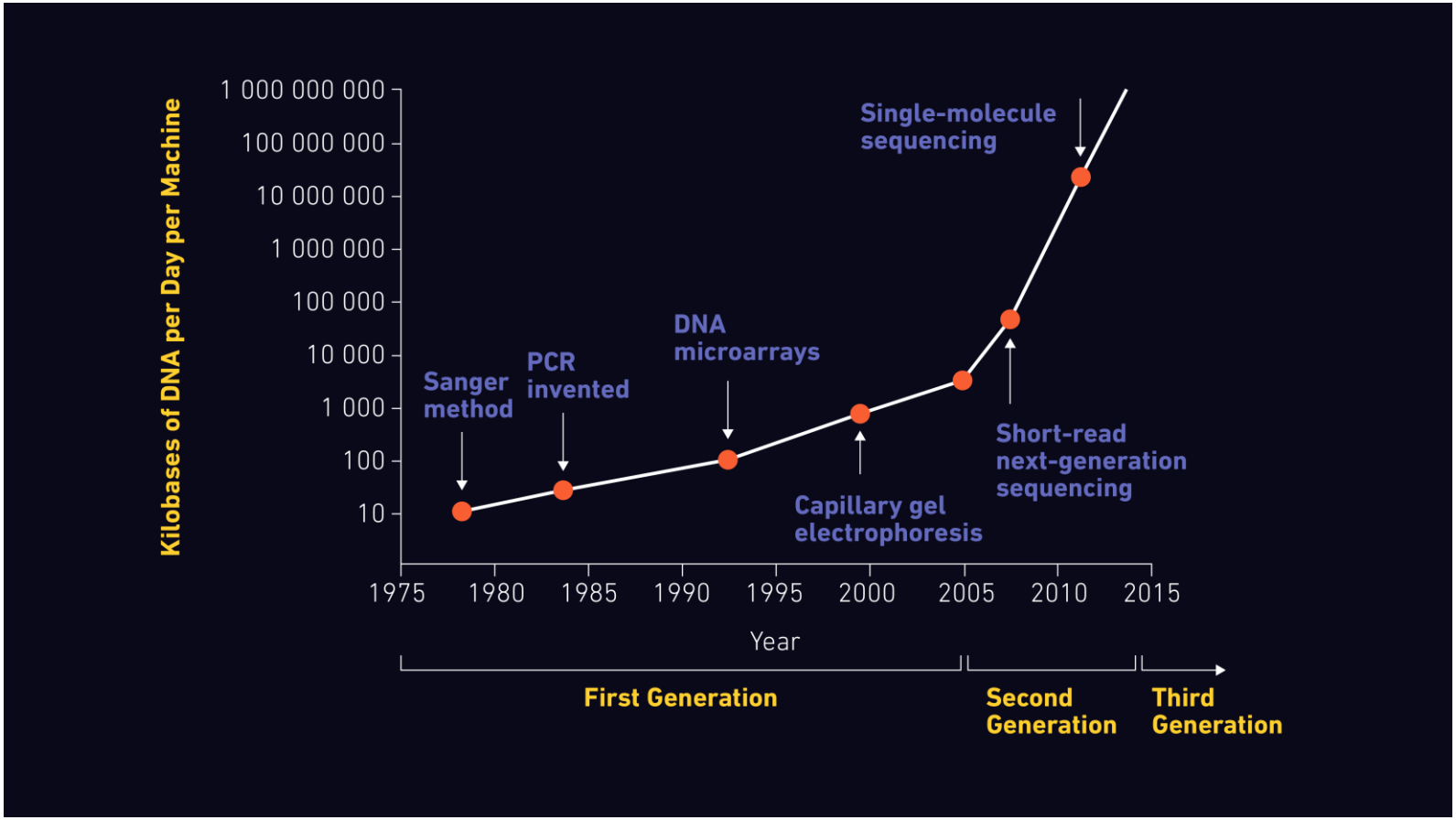
only one sequencing reaction could be analyzed at a time

Costly and time comsuming



https://en.wikipedia.org/wiki/Sanger_sequencing

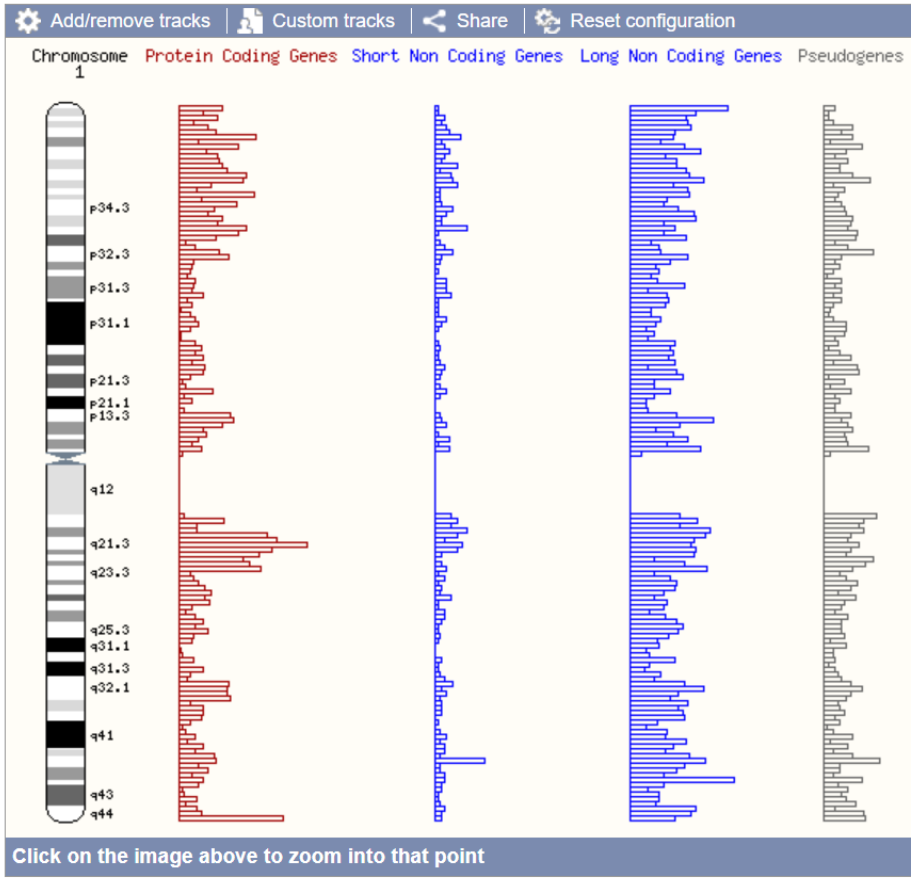
The history of sequencing

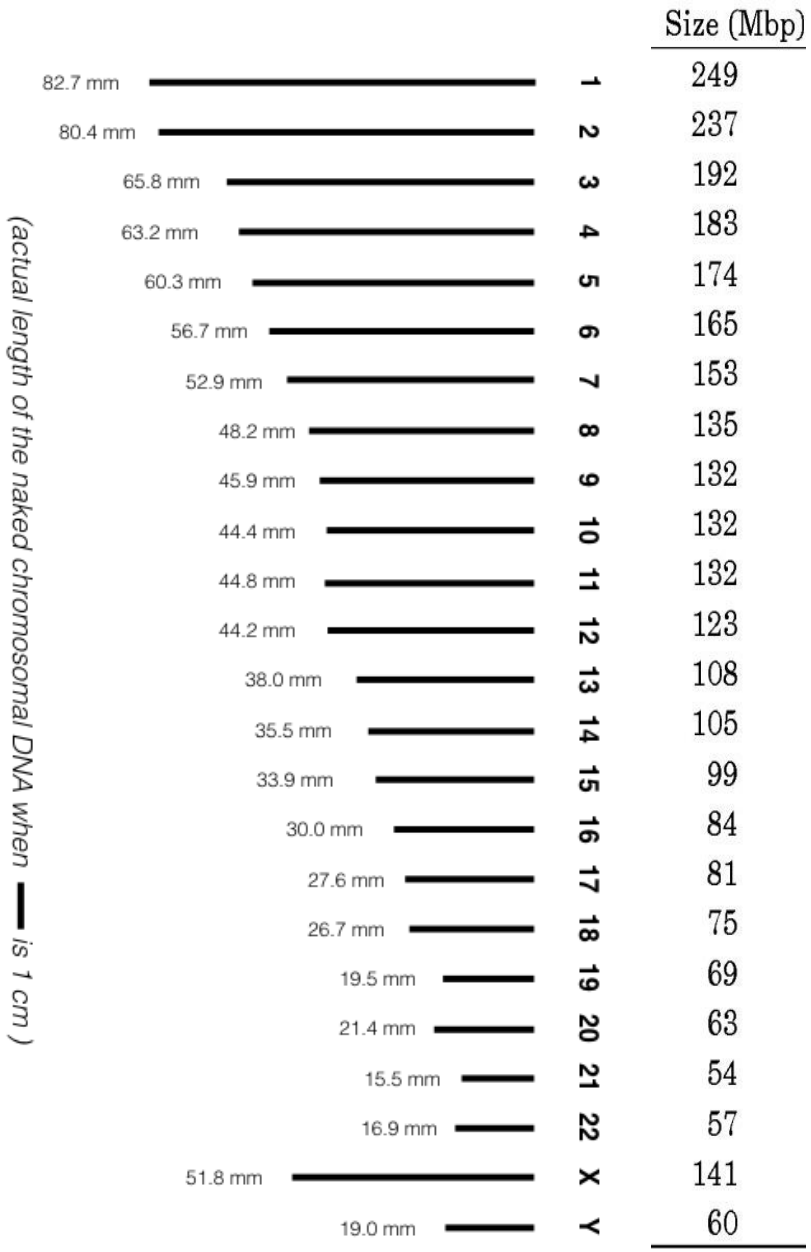


https://www.technologynetworks.com/genomics/articles/next-generation-sequencing-ngs-definition-and-overview-346532

The human genome

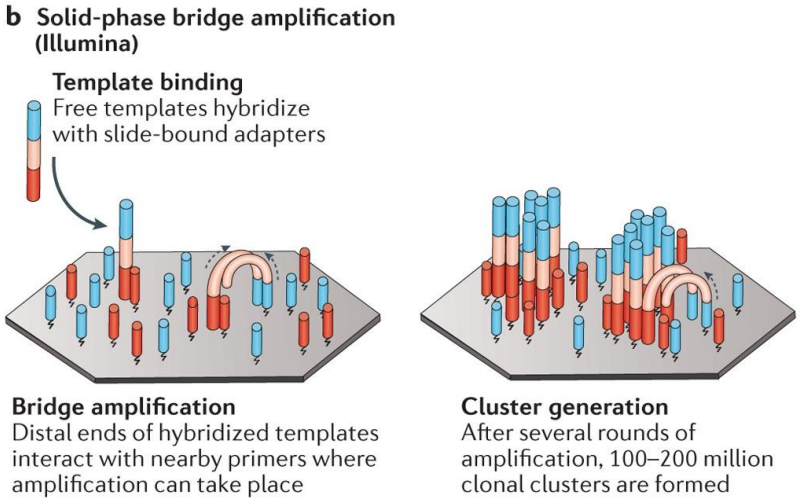
The chromosome 1





Second generation

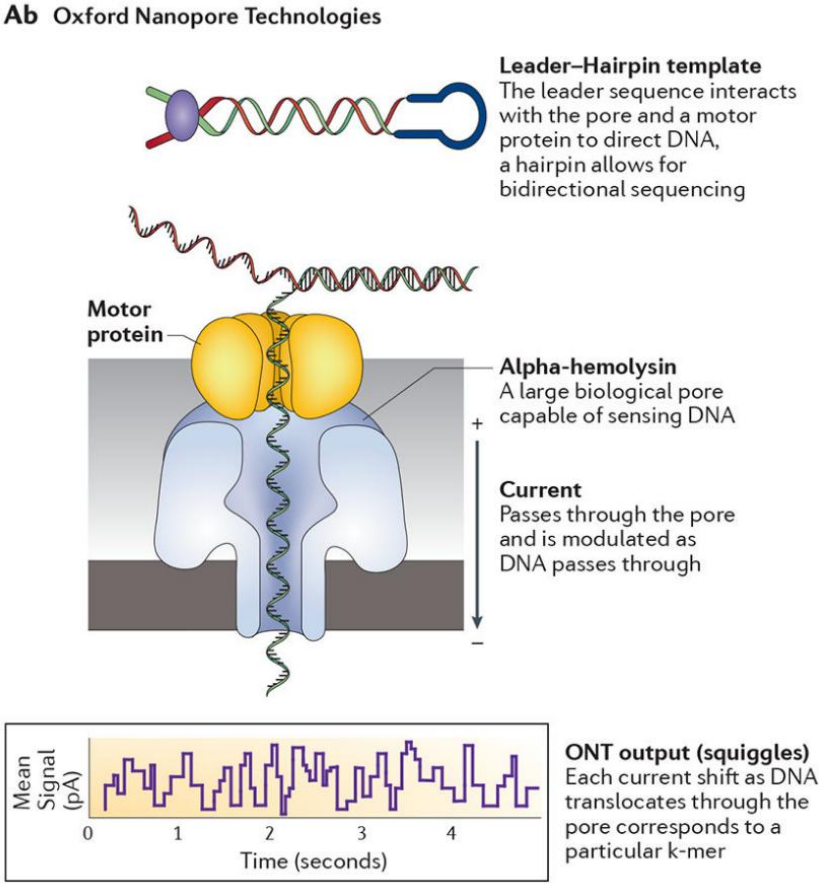
- Between 2004 and 2006 ‘‘next-generation sequencing (NGS)”
- ‘‘short-read” sequencing technologies: massively parallel sequencing, whereby millions of individual sequencing reactions occur in parallel.
- 100 - 150 bp



Illumina sequencing technology (1) Let’s watch a video

Third generation:Nanopore

- Most common PacBio and Nanopore
- Genomes are highly complex with long repetitive regions
- Long reads (10–100 kb+)
- For mRNA: entire length
- It is able to detect directly DNA and RNA modifications

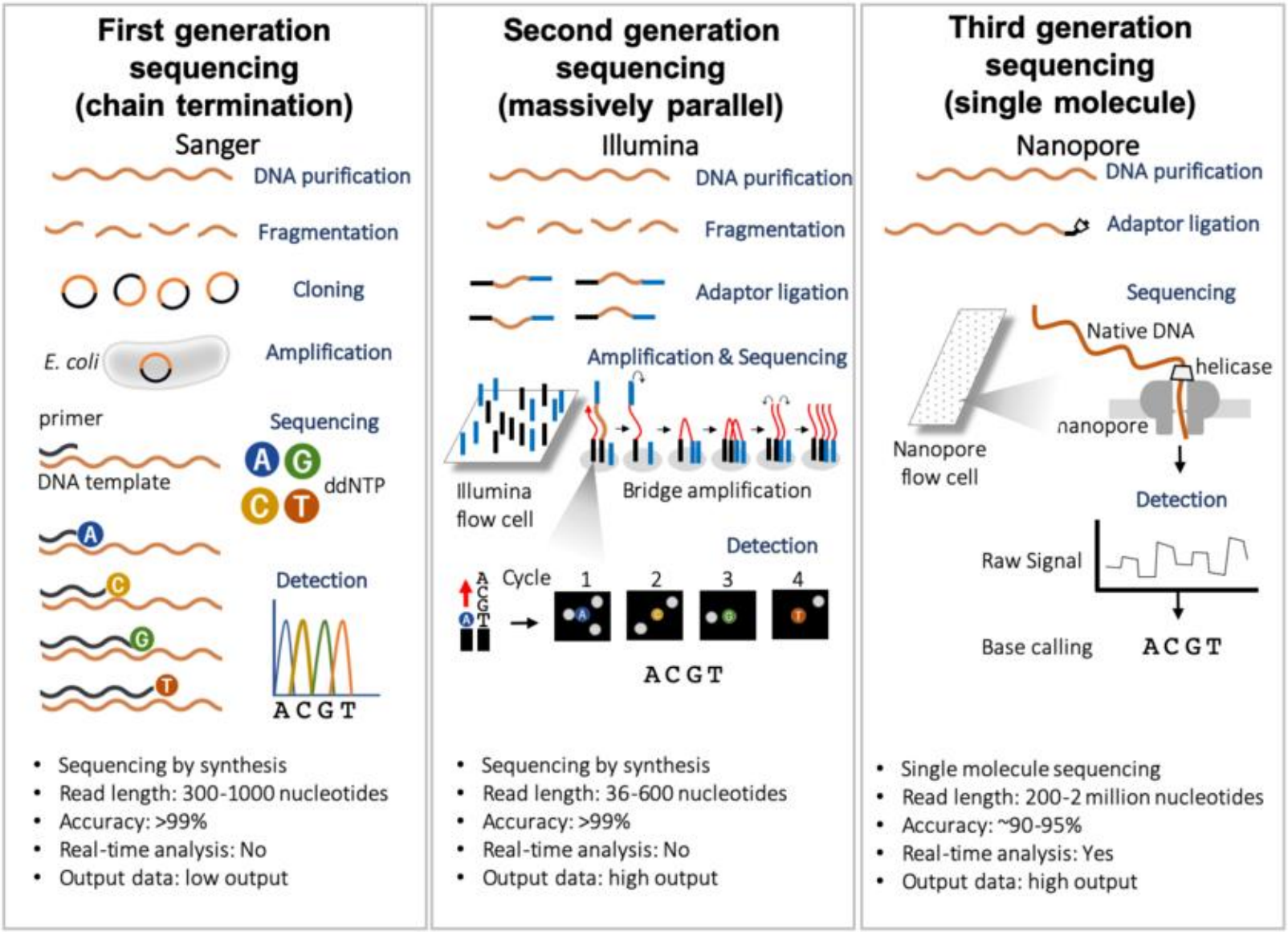


Goodwin, S., McPherson, J. D. & McCombie, W. R. Coming of age: ten years of next-generation sequencing technologies. *Nat Rev Genet* **17**, 333–351 (2016).

Oxford Nanopore

- Genome/DNA sequencing, incl. targeted sequencing
- Methylation detection:
  - DNA: 5-Methylcytosine (5mC), 5-Hydroxymethylcytosine (5hmC), N6-Methyladenine (6mA) and 4-methylcytosine (4mC)
  - RNA: N6-methyladenosine (m6A) and pseudouridine (pseU)
- Chromatin confirmation capture (3C) - Pore-C
- Long-read RNA sequencing - bulk and single-cell sequencing
- Microbial and metagenomics

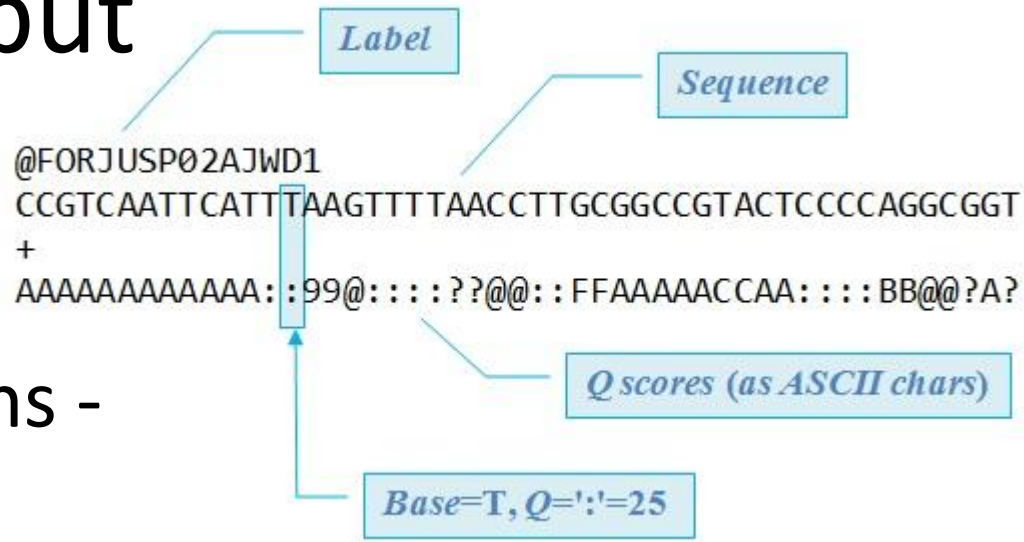
www.nanoporetech.com



Patterson, John & Wongsurawat, Thidathip & Rodriquez, Analiz. (2020).
A Glioblastoma Genomics Primer for Clinicians. Medical Research Archives. 8. 1-17. 10.18103/mra.v8i2.2034.

The world of sequencing

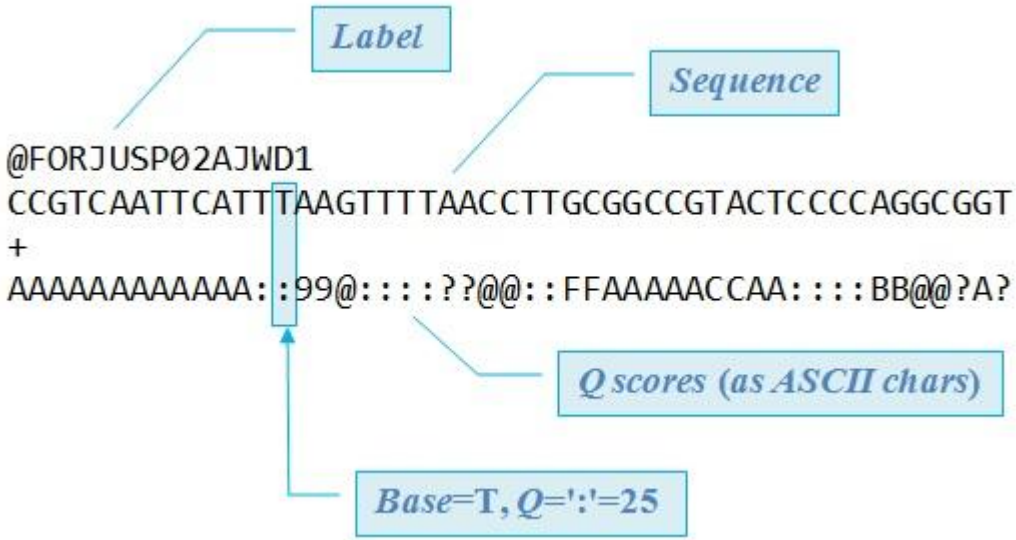
Sequencing output



- **Fastq**
- Typical file extensions - *.fastq, *.fq
- If compressed - *.fastq.gz, *.fq.bz
- If paired end then for each sample there'll be *R1.fq, *R2.fq

Fastq

- Text file for read sequences & quality scores–
  - Line 1 = '@' character, followed by information about the sequencing run (also a sequence identifier)
  - Line 2 = **raw sequence** letters
  - Line 3 = A separator, which is simply a plus (+) sign and is optionally followed by the info in Line 1
  - Line 4 = encodes the quality values for the sequence in Line 2



FASTQ Label

```
@<instrument>:<run number>:<flowcell ID>:<lane>:<tile>:<x-pos>:<y-pos> <read>:<is filtered>:<control number>:<sample number>
```

The following table describes the elements:

Element	Requirements	Description
@	@	Each sequence identifier line starts with @
<instrument>	Characters allowed: a–z, A–Z, 0–9 and underscore	Instrument ID
<run number>	Numerical	Run number on instrument
<flowcell ID>	Characters allowed: a–z, A–Z, 0–9	
<lane>	Numerical	Lane number
<tile>	Numerical	Tile number
<x_pos>	Numerical	X coordinate of cluster
<y_pos>	Numerical	Y coordinate of cluster
<read>	Numerical	Read number. 1 can be single read or Read 2 of paired-end
<is filtered>	Y or N	Y if the read is filtered (did not pass), N otherwise
<control number>	Numerical	0 when none of the control bits are on, otherwise it is an even number. On HiSeq X and NextSeq systems, control specification is not performed and this number is always 0.
<sample number>	Numerical	Sample number from sample sheet

Base Quality score (BQ)/Phred score

- Derived from Sanger sequencing
- It indicates the probability that a given base is called incorrectly by the sequencer; parameters relevant to a particular sequencing chemistry are analyzed for a large empirical data set of known accuracy and the resulting tables are used to assign BQ to new data
- Q = -10 * log10 P (P is the probability of base call error, ranging from 0 to 1)
  - If P = .1 (1E-1) (10%) Q=10
  - If P = .01 (1E-2) (1%) Q=20
  - If P = .001 (1E-3) (0.1%) Q=30
  - If P = .0001 (1E-4) (0.01%) Q=40
  - Low Q scores can lead to increased false positive variant calls, inaccurate conclusions & high experimental validation costs.

https://www.ebi.ac.uk/training/materials/genome-bioinformatics-materials/raw-read-preprocessing-qc-read-mapping/

Q Score ASCII Encoding

- The first 33 characters in the ASCII-table are unprintable control codes and are used to control peripherals such as printers
- Offset the ASCII code by 33 to get Phred score
- Beware - older read data offset by 64!

Q score/ Phred score

```
@1754ed0c-e78a-4f93-a956-249e17d2c608
runid=c821e002d29938eb71e31a54220ea3e84a43963c sampleid=01 read=1
54 start_time=2019-08-22T08:02:40Z
CAGTATGCGCCCGTTCAATCGGTGGTGTTTTATGATCATACGCCTACAGTGACGAAATTTGTCGC
TGTTTCTGTTGGTGCTGATATTGCATACGATGGCGGGTGTAGATAATAAACGGAGCCAAATTGCA
CGGAGAAGAAGACTAGAGCGACAAACCAGCGTCTAAAGACACCGACAACAGCGTA
+
)$%$$##&###%$046+,(&#$&%'7*0010,-*%%$$$$$#$$&%&$$,.10%&///7;/3;,2
$$$%%$4731/+24'+.,16<>3/./'%$$$$#&)%'(%-31+-&',,&'&+,*/$&&&%(&)**
35713464532%&$,---%(-%%0/('&')&&%$#&*(,$00((/0&$%#$%%$#&'.1/$
@a06a565e-bc77-4821-a7ea-e0571c203a38
runid=c821e002d29938eb71e31a54220ea3e84a43963c sampleid=01 read=1
54 start_time=2019-08-22T08:03:08Z
CTGGTGTTAGCTGGTTCATTTTATCAAGTTTATACCGTGACAAAGAAAGTTGTCGGTGTCTGTGA
CTTCTCCCTCCTTGTTGATATTACTTCGCGCGATAAATCTGCTCACCGGGGAGTGTCGCGCTGCA
GGCGAACTTTAAAGACGCCAGACGGCACTGTACGT
+
)$#%$$)+$&%('*&$&-,-.,%*+#$$$$$$%%,.+:98:89)&2/-415%.2%,4,/./,)*$
=<=A967*2'/'0'.&*-)*0&#$$#%%()+&%&%-*+)$$$$%)1%&)#'%%&(,*+,$$#$%#
(,#$%&$$$$$$$&$&$)%&&%&$'#$&#%'%%&'%%
```

) = 41

41 – 33 = 8

$P = 10^{-Q/10}$

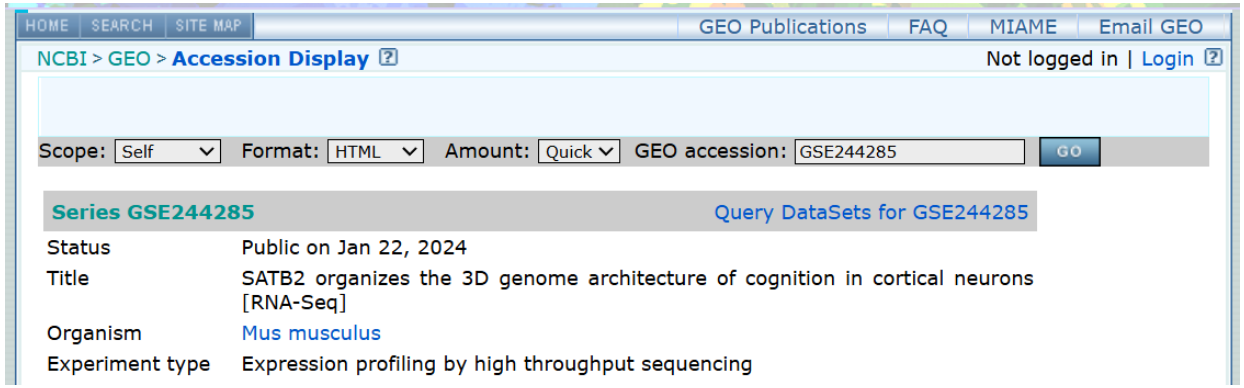
Where P is the **Probability of Incorrect Base Call**

DEC	Symbol	Description
32	SP	Space
33	!	Exclamation mark
34	"	Double quotes (or speech marks)
35	#	Number sign
36	$	Dollar
37	%	Per cent sign
38	&	Ampersand
39	'	Single quote
40	(	Open parenthesis (or open bracket)
41	)	Close parenthesis (or close bracket)
42	*	Asterisk
43	+	Plus
44	,	Comma
45	-	Hyphen-minus
46	.	Period, dot or full stop
47	/	Slash or divide
48	0	Zero
49	1	One
50	2	Two
51	3	Three
52	4	Four
53	5	Five
54	6	Six
55	7	Seven
56	8	Eight
57	9	Nine
58	:	Colon
59	;	Semicolon

60	<	Less than (or open angled bracket)
61	=	Equals
62	>	Greater than (or close angled bracket)
63	?	Question mark
64	@	At sign
65	A	Uppercase A
66	B	Uppercase B
67	C	Uppercase C
68	D	Uppercase D
69	E	Uppercase E
70	F	Uppercase F
71	G	Uppercase G
72	H	Uppercase H
73	I	Uppercase I
74	J	Uppercase J
75	K	Uppercase K
76	L	Uppercase L
77	M	Uppercase M
78	N	Uppercase N
79	O	Uppercase O

Where to find FASTQ

- Major database: GEO
- Data stored in SRA format (highly compressed file)
- To get the FASTQ file: SRA-Toolkit (fasterq-dump)

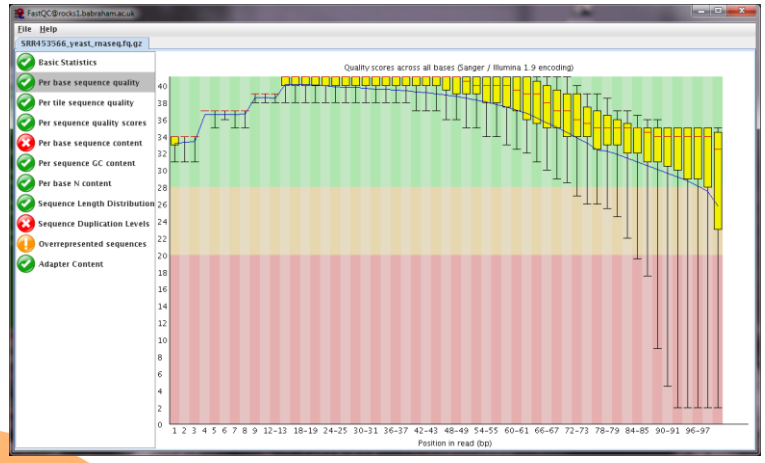


Read Pre-Processing

Raw reads (FASTQ)

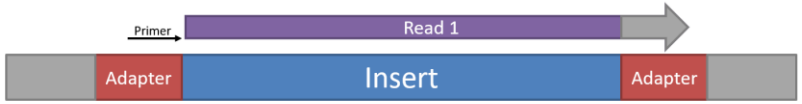
Check quality

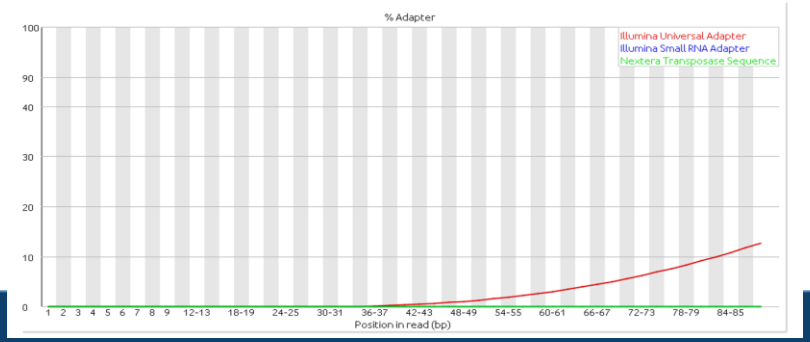
FastQC



Trim/remove unwanted reads

Cutadapt/ Trimmomatic



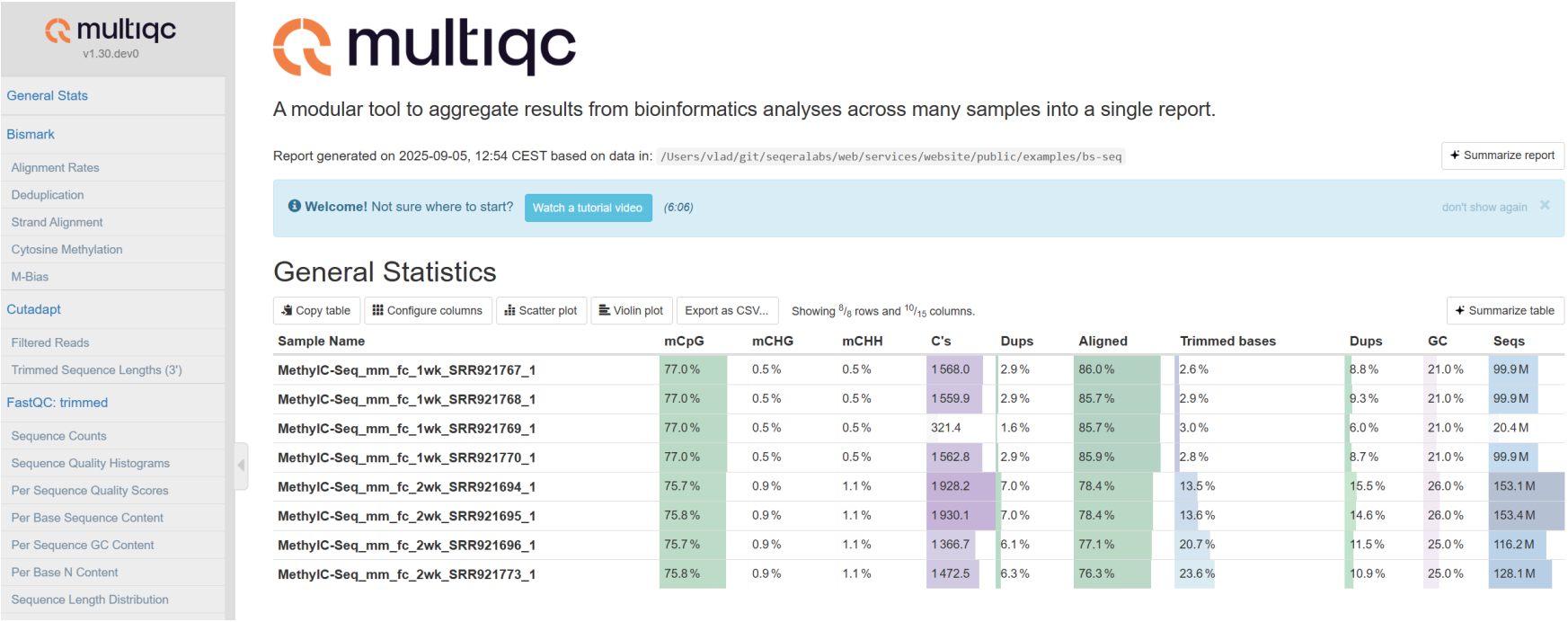


Cleaned reads (FASTQ)

Check quality

FastQC

MultiQC



Sample Name	mCpG	mCHG	mCHH	C's	Dups	Aligned	Trimmed bases	Dups	GC	Seqs
MethyIC-Seq_mm_fc_1wk_SRR921767_1	77.0 %	0.5 %	0.5 %	1 568.0	2.9 %	86.0 %	2.6 %	8.8 %	21.0 %	99.9 M
MethyIC-Seq_mm_fc_1wk_SRR921768_1	77.0 %	0.5 %	0.5 %	1 559.9	2.9 %	85.7 %	2.9 %	9.3 %	21.0 %	99.9 M
MethyIC-Seq_mm_fc_1wk_SRR921769_1	77.0 %	0.5 %	0.5 %	321.4	1.6 %	85.7 %	3.0 %	6.0 %	21.0 %	20.4 M
MethyIC-Seq_mm_fc_1wk_SRR921770_1	77.0 %	0.5 %	0.5 %	1 562.8	2.9 %	85.9 %	2.8 %	8.7 %	21.0 %	99.9 M
MethyIC-Seq_mm_fc_2wk_SRR921694_1	75.7 %	0.9 %	1.1 %	1 928.2	7.0 %	78.4 %	13.5 %	15.5 %	26.0 %	153.1 M
MethyIC-Seq_mm_fc_2wk_SRR921695_1	75.8 %	0.9 %	1.1 %	1 930.1	7.0 %	78.4 %	13.6 %	14.6 %	26.0 %	153.4 M
MethyIC-Seq_mm_fc_2wk_SRR921696_1	75.7 %	0.9 %	1.1 %	1 366.7	6.1 %	77.1 %	20.7 %	11.5 %	25.0 %	116.2 M
MethyIC-Seq_mm_fc_2wk_SRR921773_1	75.8 %	0.9 %	1.1 %	1 472.5	6.3 %	76.3 %	23.6 %	10.9 %	25.0 %	128.1 M

- Very useful and versatile tool
- It can summaries a variety of tools outputs

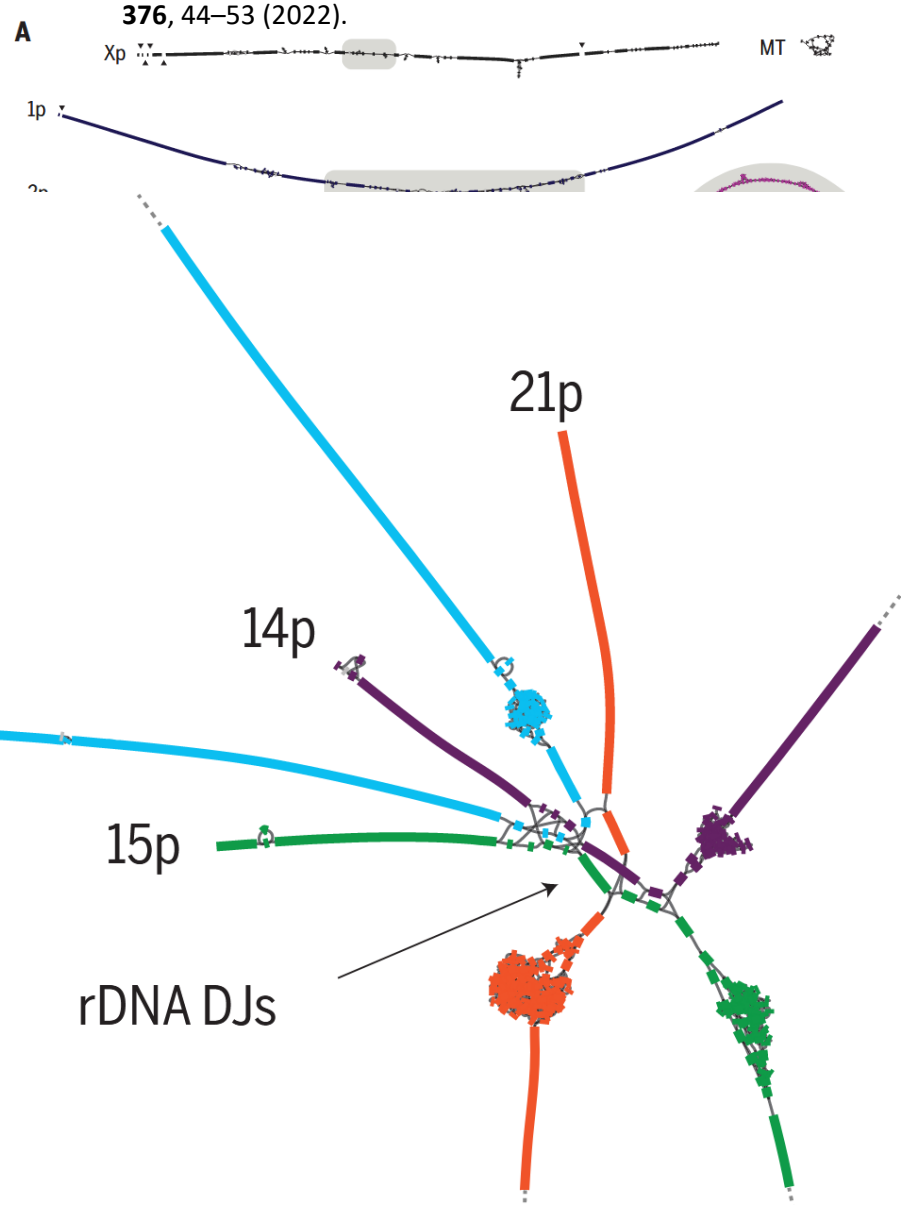
The two main sequencing

- **DNA**
  - Targeted Sequencing
    - Small, specific regions of the genome
  - Whole-Exome
    - Mutations in coding regions
  - Whole-Genome Sequencing
    - **Genome reference**
    - Evolution
    - Structural variant
    - Complex disease studies
  - Single-Cell Sequencing
    - Cell lineage tracing.
    - Developmental biology

- **RNA**
  - Targeted RNA Sequencing
    - Clinical diagnostics
    - Gene expression of specific genes
  - Bulk RNA-seq
    - Gene/transcripts expression across conditions
    - Alternative splicing research
  - scRNA-seq
    - DEA on specific cell type
    - Cell development studies
  - Spatial Transcriptomics
    - Tumor microenvironment mapping
    - Organ/tissue architecture studies
  - Other RNAs

Genome Assembly

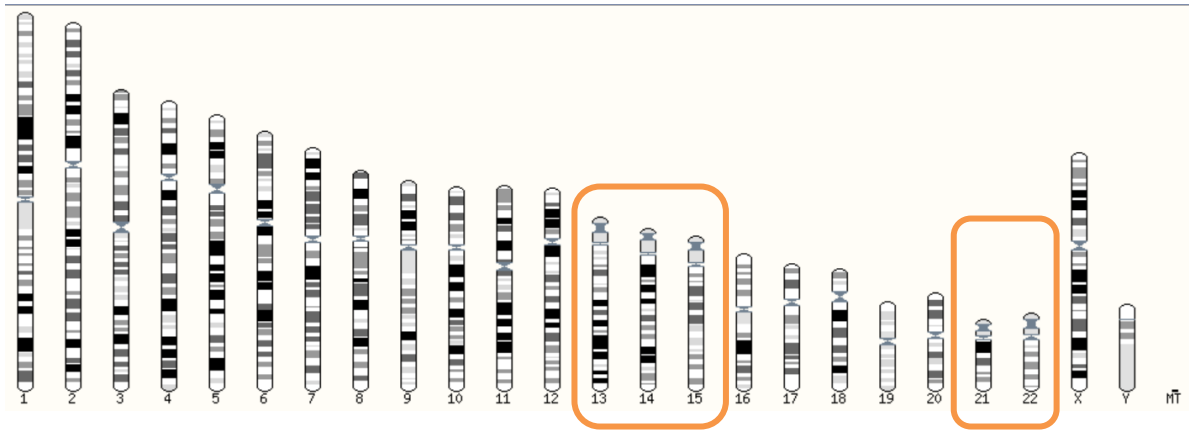
Nurk, S. *et al.* The complete sequence of a human genome. *Science* **376**, 44–53 (2022).

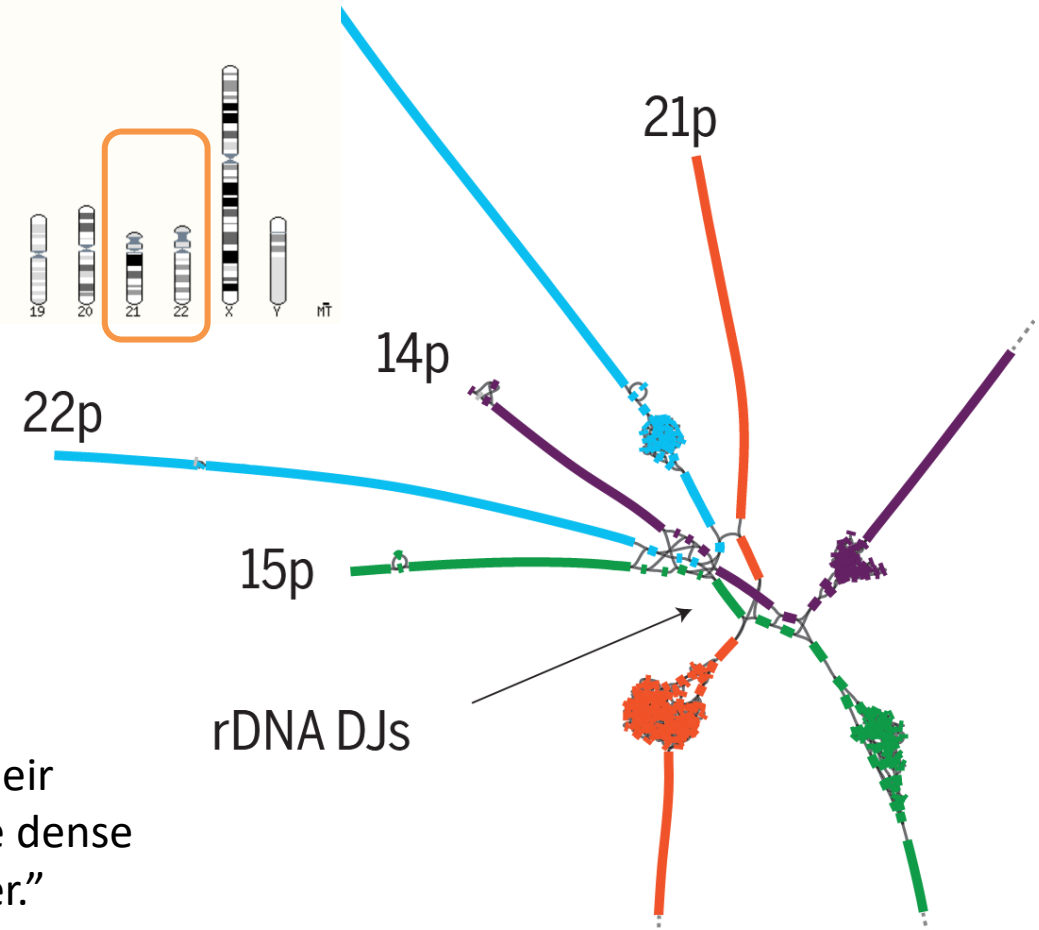


- Why is so important?
  - Reference
  - Evolution
  - Complex diseases
- Why is so difficult?
  - Highly repetitive regions

Nurk, S. *et al.* The complete sequence of a human genome. *Science* **376**, 44–53 (2022).

Genome Assembly





"The five acrocentric chromosomes are connected owing to similarity between their short arms, and the rDNA arrays form five dense tangles because of their high copy number."

Human genome current release

- GRCh38.14

Ensembl BLAST/BLAT | VEP | Tools | BioMart | Downloads | Help & Docs | Blog

Login/Register

Search Human…

Human (GRCh38.p14) ▾

Human assembly and gene annotation

Assembly

This site provides a data set based on the December 2013 *Homo sapiens* high coverage assembly GRCh38 from the Genome Reference Consortium. This assembly is used by UCSC to create their hg38 database. The data set consists of gene models built from the genewise alignments of the human proteome as well as from alignments of human cDNAs using the cDNA2genome model of exonerate.

This release of the assembly has the following properties:

- contig length total 3.4 Gb.
- chromosome length total 3.1 Gb (excluding haplotypes).

It also includes 261 alt loci scaffolds, mainly in the LRC/KIR complex on chromosome 19 (35 alternate sequence representations) and the MHC region on chromosome 6 (7 alternate sequence representations).

Watch a video on YouTube about patches and haplotypes in the Human genome.

Patches

As the GRC maintains and improves the assembly, patches are being introduced. Currently, assembly patches are of two types:

Statistics

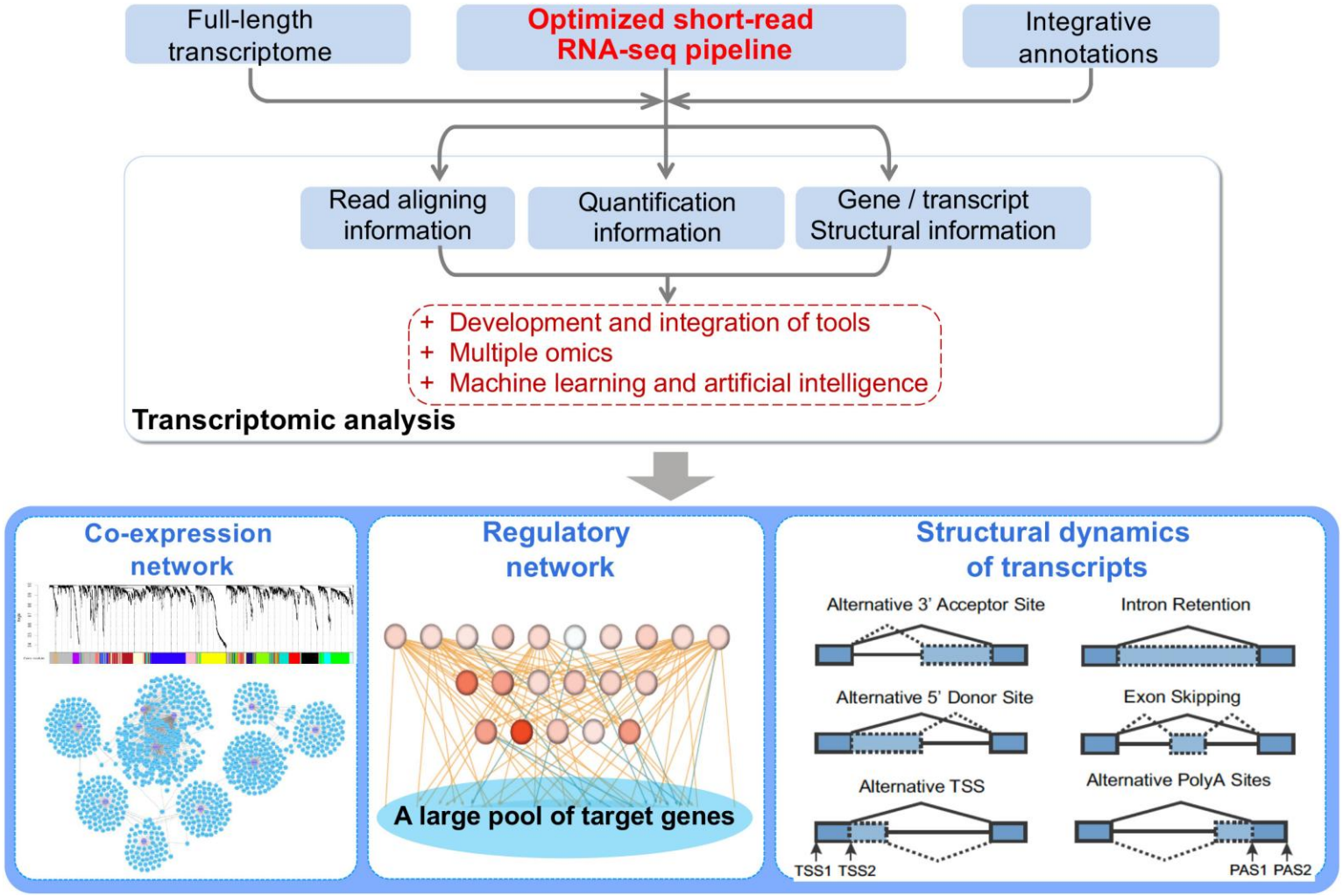
Summary

Assembly	GRCh38.p14 (Genome Reference Consortium Human Build 38), INSDC Assembly GCA_000001405.29, Dec 2013
Base Pairs	3,099,750,718
Golden Path Length	3,099,750,718
Assembly provider	Genome Reference Consortium
Annotation provider	Ensembl
Annotation method	Full genebuild
Genebuild started	Jan 2014
Genebuild released	Jul 2014
Genebuild last updated/patched	May 2025
Database version	115.38
Gencode version	GENCODE 49

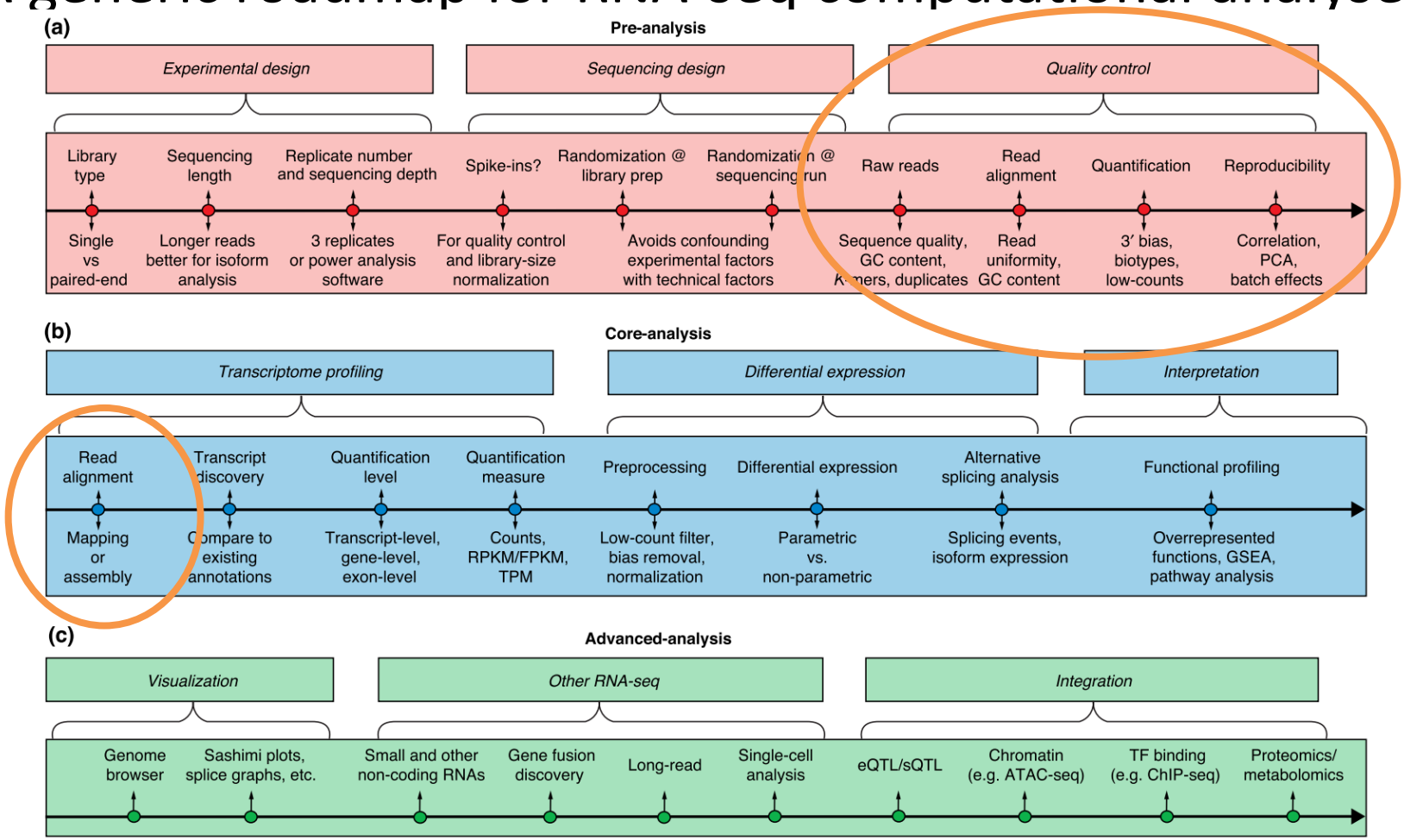
Recap some definitions

- **Gene:** sequence of nucleotides that encodes a functional molecule (mRNA, tRNA, rRNA, among others)
- **Genome:** the entire genetic information an organism (or virus) possesses
- **Transcript:** RNA molecule derived from a gene
- **Transcriptome:** The entire repertoire of transcripts present in an organism at a given time
- **Isoform:** transcript that originates from alternative transcription start sites, or alternative transcription termination sites, or by alternative splicing sites, or any combination of the above

Overview transcriptomics

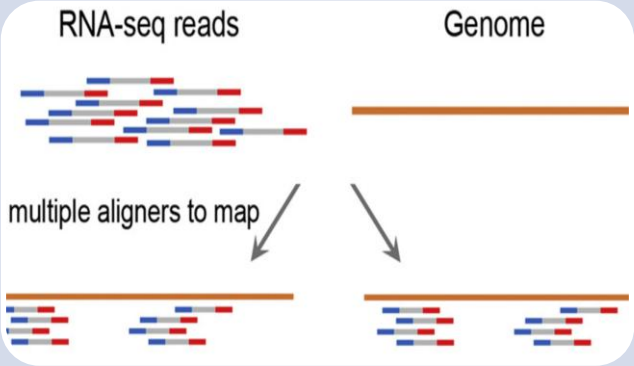


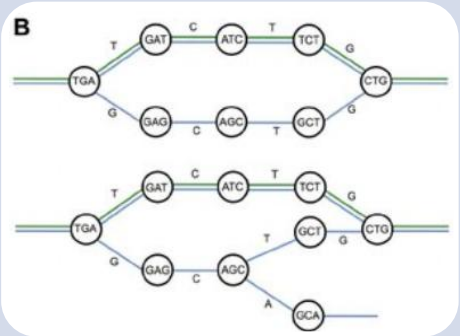
A generic roadmap for RNA-seq computational analyses



Conesa, A., Madrigal, P., Tarazona, S. *et al.* A survey of best practices for RNA-seq data analysis. *Genome Biol* **17**, 13 (2016). https://doi.org/10.1186/s13059-016-0881-8

Transcript assembly strategies





Genome-guided

Reads can be aligned/mapped to a reference genome.

If working with eukaryotes make sure to use a splice-aware aligner

De-novo

De Bruijn Graphs can be used to assemble reads in contiguous sequences (contigs).

Such assemblers assume an even distribution of a finite amount of k-mers

Questions?